



A Comparative Methodological Framework for Evaluating Distributional and Nutritional Effects of Food Carbon Pricing

Ghassan Al-Masbhi^{1*}, Cathal O'Donoghue¹, Zeynep Gizem Can¹, Anna Diop-Christensen², Ana Montes³, Emily Paradowska³, Martina Pezer⁴, Denisa Sologon³

¹University of Galway; BiOrbic SFI Research Centre, Galway, Ireland; ²University College of Northern Denmark, Hjørring, Denmark; ³Luxembourg Institute of Socio-Economic Research, Esch-sur-Alzette, Luxembourg; ⁴Institute of Public Finance, Zagreb, Croatia

Abstract Food systems contribute significantly to greenhouse gas emissions, but carbon pricing in this sector remains limited and its distributional implications are not well understood. This paper develops a cross-country microsimulation framework that links Life Cycle Assessment (LCA) emissions intensities with Household Budget Survey (HBS) microdata to assess a food carbon tax in seven EU Member States and Türkiye. The model applies a uniform carbon tax to emissions embodied in food purchases, assumes full pass through to consumer prices and holds household quantities fixed. It produces household level outcomes on food expenditure, diet related emissions, caloric cost per kilojoule and the tax burden as a share of income. Results point to small but consistent regressive effects: post tax Gini coefficients increase slightly in all countries, Reynolds–Smolensky indices are negative and Kakwani values indicate a higher relative burden on lower income groups. A sensitivity analysis using alternative EU and global LCA datasets shows that, despite variation in some emission factors, the relative ordering of foods and the distributional patterns remain broadly similar. The framework can be further extended to incorporate behavioural responses, alternative tax designs and broader food system policies.

DOI: <https://doi.org/10.34196/ijm.00336>

***For correspondence:**

G.Al-Masbhi1@
universityofgalway.ie

©This article is distributed under the terms of the [Creative Commons Attribution License](#), which permits unrestricted use and redistribution provided that the original author and source are credited.

Author Keywords:

Microsimulation, Food carbon taxation, Life Cycle Assessment (LCA), Distributional analysis
© 2026, Al-Masbhi et al.

1. Introduction

Modelling the distributional impacts of food carbon pricing is important for understanding how climate policies affect households. The agri-food system contributes nearly one third of global greenhouse gas (GHG) emissions, with livestock and fertiliser use driving most of this footprint (Crippa et al., 2021). In the EU, agriculture accounts for about 10% of emissions, yet carbon pricing remains focused on energy and industry (Ramstein, 2019). Food systems are often excluded because dietary habits, cultural norms and socio-economic conditions vary widely across countries (Smith and Gregory, 2013). These cross-country differences, ranging from meat heavy diets in Western Europe to higher food budget shares in Eastern Europe, influence how carbon taxes are transmitted to households (Dokova et al., 2022). This paper introduces a microsimulation framework to assess household level impacts of food carbon taxes across eight countries, integrating expenditure, price and emissions data to explore trade-offs between climate objectives and distributional outcomes.

A substantial literature has examined fiscal instruments in food and environmental policy, including health motivated taxes, subsidies and carbon pricing initiatives. Public health studies have analysed the nutritional and expenditure effects of taxes on unhealthy foods or beverages (*Segal et al., 2022; Smed et al., 2016; Tiffin et al., 2015*), while environmental economics has increasingly focused on carbon pricing as a mitigation tool, including in agriculture (*Kebreab et al., 2022; Sørensen et al., 2025*). A range of modelling approaches reflects this diversity of objectives. Macroeconomic frameworks such as Computable General Equilibrium (CGE) and Input Output (IO) models capture system wide interactions (*Böhringer et al., 2017; Ntombela et al., 2019*). Demand system models such as AIDS, QUAIDS and EASI simulate price driven changes in consumption (*Caillavet et al., 2019; García-Muros et al., 2017*). Agent based models examine dynamic system responses (*Roxburgh et al., 2025*) and microsimulation models use household level microdata to assess tax incidence and food expenditure patterns (*Callan et al., 2009; O'Donoghue, 2014*). While each strand contributes useful insights, relatively few studies bring together environmental, nutritional and equity dimensions within a comparable framework across multiple countries.

In a European policy context, cross country differences may be relevant for both the level and the design of food carbon taxation. Policy instruments are increasingly being discussed at the European level as well as within individual Member States, and any coordinated carbon tax is likely to interact with national income distributions, food budget shares and dietary patterns, which can lead to different distributional outcomes even under the same tax rate (*Büchs and Schnepf, 2013; Callan et al., 2009*). Studies of energy related carbon taxes suggest that countries with similar headline measures of regressivity can reach these outcomes through different underlying mechanisms, which indicates that a uniform European tax might have uneven welfare effects if national structures are not fully considered (*Dorband et al., 2019; Klenert et al., 2018*). A better understanding of these cross-country differences can therefore help in thinking about policy tools, such as targeted subsidies or revenue recycling schemes, that could alleviate regressive effects while remaining consistent with a broader European climate framework (*Sterner, 2012; Varacca et al., 2024*).

Despite progress, several gaps remain. Cross-country analyses of food carbon taxation are still scarce, as most studies focus on single-country settings (*Varacca et al., 2024*). Several modelling approaches concentrate mainly on environmental or economic outcomes, and only a smaller group of contributions considers nutrition, affordability and emissions jointly within one framework (*Bonnet et al., 2018; Säll and Gren, 2015; Zhang et al., 2024*). When using HBS-based microsimulation to assess equity, a key challenge is the prevalence of zero expenditures due to short recall periods, infrequent purchases and underreporting. If not explicitly addressed, these zeros can bias elasticity estimates and complicate behavioural modelling (*Deaton and Irish, 1984; Shonkwiler and Yen, 1999; Peltner and Thiele, 2021*). Existing HBS based microsimulation studies often focus on single countries, do not always link food demand to emissions data and only partially account for interactions across food groups, so the links between emissions reduction, nutrition and affordability are not yet fully explored (*Al-Masbhi et al., 2026; Edjabou and Smed, 2013; Forero-Cantor et al., 2020; Garnett, 2011; Maestre-Andrés et al., 2019; Revoredo-Giha et al., 2018*).

These considerations motivate the research questions explored in this paper. First, how does a harmonised food carbon tax affect household food expenditure, diet-related emissions and caloric intake across countries? Second, how do these effects vary across national contexts with distinct dietary patterns, production systems and income distributions? Third, what trade-offs arise between emission reduction, distributional impacts and dietary energy in a comparable tax setting? Examining these questions helps clarify why household-level impacts differ across countries and why national structures matter for the design of food carbon pricing.

This paper develops a cross-country microsimulation framework that links Life Cycle Assessment (LCA) emissions data with Household Budget Survey (HBS) information on food expenditure and caloric content for eight countries: Ireland, France, Spain, Denmark, Poland, Luxembourg, the United Kingdom and Türkiye. By combining emissions intensities (kg CO₂e/kg) nutritional information and household spending patterns, the framework evaluates the distributional and dietary implications of

applying a carbon tax to food.¹ The analysis focuses on household-level outcomes food expenditure, diet-related emissions, the cost of dietary energy and the tax burden as a share of income under fixed quantities, without behavioural responses.

Methodologically, the study employs a static microsimulation model that applies a carbon tax to the emissions embodied in food items, assumes full pass-through into retail prices, recalculates post-tax prices and traces changes in household expenditure, tax payments and diet-related emissions across income deciles and countries. The model does not rely on EUROMOD and does not incorporate substitution effects, behavioural elasticities or interactions with benefit systems. Distributional outcomes are summarised using standard inequality and progressivity measures, including pre- and post-tax Gini coefficients, the Kakwani index and the Reynolds–Smolensky index. A sensitivity analysis using alternative LCA datasets tests how assumptions on emissions intensities influence results.

The eight countries in this study (Ireland, France, Spain, Denmark, Poland, Luxembourg, the UK, and Türkiye) offer a varied and policy-relevant context for evaluating food carbon taxation. Empirical studies from Ireland, France, Denmark and the UK show that diet-related greenhouse gas emissions are largely driven by ruminant meat, dairy products and discretionary processed foods, with higher per-capita dietary emissions in central and western European countries than in Mediterranean settings (*Alves et al., 2024; Hyland et al., 2017; Murakami and Livingstone, 2018; Rippin et al., 2021; Trolle et al., 2022; Vieux et al., 2012*). Poland and Türkiye rely more on cereals, legumes, and vegetables, which lowers average dietary emissions but increases exposure to food price shocks, since low income households spend a larger share of their budgets on food (*Gumus et al., 2010; Rejman et al., 2014*). Higher consumption of convenience and packaged foods in Ireland and the UK involves more energy-intensive processing, refrigeration and packaging, increasing upstream emissions (*Browne et al., 2023; Garnett, 2011*). By contrast, Poland and Türkiye make greater use of seasonal produce and shorter domestic supply chains, which can lower transport emissions but heighten vulnerability to climate and market shocks for fresh foods (*Aysoy et al., 2015; Drejerska and Sobczak-Malitka, 2023*). These contrasts in diets, supply chains and food budget shares make this set of countries well suited to studying how a uniform food carbon tax translates into different distributional and nutritional outcomes.

The remainder of this paper is structured as follows: Section 2 describes the data sources, emissions accounting, and microsimulation model. Section 3 presents the cross-country results and Section 4 concludes and offers directions for future research.

2. Methodology and Data

Developing fair and effective carbon pricing in the food sector requires understanding how dietary habits, emissions and income interact. This analysis uses microsimulation to link household consumption with emissions and nutritional outcomes across different carbon tax scenarios (*Figari et al., 2015; O'Donoghue, 2014*).

2.1. Microsimulation

We develop and use a static, non-behavioural microsimulation model to quantify the distributional and nutritional effects of a food carbon tax. The model applies policy changes to household-level microdata while preserving heterogeneity in income, consumption and demographics (*Figari et al., 2015; O'Donoghue, 2014; Tanton, 2014*). Microsimulation has been used to analyse emissions trading, agricultural emission taxes and carbon pricing schemes (*Cervigni et al., 2013; Hynes et al., 2009; Wadud et al., 2008*).

To link consumption with nutrition, we use nutrient information from supermarket labels, complemented by the EFSA Food Composition Database. Energy values (kJ per 100 g) are assigned to each HBS food item after converting expenditures into quantities using country-specific average prices and aligning products with the LCA classification. Each item therefore has both an emissions intensity and

1. kg CO₂e per kg means kilograms of carbon-dioxide equivalent emissions generated for each kilogram of product produced.

an energy content. Household energy intake is obtained by multiplying quantities by these energy values, which allows us to compare dietary energy, its cost and the carbon intensity of diets (CO₂e per kilojoule) across countries.

The incidence analysis is conducted under standard first round assumptions. Household quantities are held fixed and the per unit carbon tax is assumed to be fully passed through into retail prices. These assumptions are commonly used in distributional microsimulation of indirect taxes, where the focus is on statutory tax burdens rather than behavioural or general equilibrium feedbacks (*Decoster et al., 2010*), and they are also adopted in recent work on food related carbon pricing (*Ricci et al., 2024*). Within this framework, the model traces how a given carbon tax affects food expenditure, tax payments, diet related emissions and the cost of dietary energy across the income distribution, without modelling substitution between foods or broader market adjustments.

Conceptually, the model follows a simple sequence. A food carbon tax raises the consumer prices of more emission intensive products relative to lower emission alternatives. Given observed consumption patterns, this changes the share of total food spending that is associated with high carbon items and the share of income absorbed by food and by the tax. These shifts in spending and tax incidence, combined with information on diet related emissions and calorie costs, provide the basis for evaluating the distributional consequences of the tax and its implications for nutritional indicators in the simulated setting.

2.2. Consumption

Household food consumption patterns in the EU are shaped by a combination of disposable income, household composition, and broader structural and market dynamics (*Reisch et al., 2013*). At a basic level, food consumption is determined by the income remaining after savings. Let Y_i represent total household income and S_i the household-specific savings rate. Then, overall consumption C_i can be expressed as:

$$C_i = Y_i \cdot (1 - S_i) \quad (1)$$

A fraction of this is allocated to food, denoted $\theta_{\{f,i\}}$, such that:

$$C_{\{f,i\}} = Y_i \cdot (1 - S_i) \cdot \theta_{\{f,i\}} \quad (2)$$

In line with Engel's Law, the share of income spent on food typically decreases as income rises, making lower-income households more vulnerable to food price fluctuations and to fiscal policies like carbon taxation (*Caillavet et al., 2016*). This income sensitivity remains especially high in lower-income EU countries, where food accounts for a larger share of household expenditure among the lowest deciles, compared to higher-income countries, where the food budget share is smaller despite higher absolute spending levels (*Eurostat, 2020*).

2.3. Emissions Measurement Using Life Cycle Assessment (LCA)

Methodological choices in emissions accounting are not neutral; they shape the outcomes of distributional tax analysis. While EE-IO models capture broad economic linkages, they often aggregate food categories and obscure nutritional distinctions. In contrast, (LCA) offers food-level specificity, making it more suitable for studies that integrate environmental and dietary analysis (*Ivanova et al., 2017; Poore and Nemecek, 2018*). Prior research suggests that LCA-based taxation may overburden rural, livestock-dependent households (*Clune et al., 2017*), while IO models may underestimate vulnerabilities in urban, processed-food-heavy diets (*Chaudhary et al., 2018; Forero-Cantor et al., 2020*). However, recent evidence from *Al-Masbhi et al. (2026)* indicates that, despite these structural differences, distributional inequality results remain broadly consistent across methods. Building on this, we adopt LCA to take advantage of its finer resolution, which allows us to model nutrient-emission trade-offs and simulate targeted policy instruments with greater precision.

In this study, each food item is assigned a country-specific LCA emission factor, harmonised across products and countries (details in Section 2.6). Total dietary emissions for household h are:

$$CO_2e_h = \sum_i Q_{ih} \times EF_i^c \quad (3)$$

where Q_{ih} is the quantity of food item i consumed by household h ,

EF_i^c is the country-specific LCA emissions factor for food item i , expressed in kg CO₂e/kg.

Quantities Q_{ih} are derived by converting expenditure into physical amounts using national average prices:

$$Q_{ih} = \frac{E_{ih}}{P_i^c} \quad (4)$$

To maintain comparability, prices and emission factors are country-specific, while the structure of food items and categories is harmonized across the dataset.

All emission factors are expressed in carbon dioxide equivalents (CO₂e) using IPCC AR5 100-year Global Warming Potentials (GWPs):

$$EF_i^c = CO_{2,i} + 28 \cdot CH_{4,i} + 265 \cdot N_{2}O_{,i} \quad (5)$$

This conversion accounts for the higher warming potential of methane (CH₄) and nitrous oxide (N₂O), which are central to livestock and fertiliser-intensive systems. Using country-specific emission coefficients within a harmonised LCA framework links household consumption to total and nutrient-specific emissions and enables cross-country comparison of the equity and nutritional effects of food carbon taxation.

Emissions intensity is central to how a food carbon tax affects different products and income groups. In our LCA data, cross country variation in emission factors reflects differences in farming systems and supply chains documented in earlier work (*Clune et al., 2017; Crippa et al., 2021; Poore and Nemecek, 2018*). For ruminant meat and dairy, country level coefficients capture variation in feed composition, grazing versus housed systems, fertiliser and manure management and energy use in processing (*Notarnicola et al., 2017*). For fruits and vegetables, differences arise more from storage, refrigeration and transport distances, including the role of imports in meeting domestic demand (*Cerutti et al., 2014; González et al., 2011*). Processed foods and beverages combine upstream emissions from multiple ingredients with energy use in manufacturing and packaging, so their emission factors depend on both product formulation and the structure of national food industries (*Blonk et al., 2008; Notarnicola et al., 2017*). In the sensitivity analysis, staples such as grains, bread and milk show modest variation across countries, whereas fruits and vegetables, meat products and ready-made foods display much wider ranges, consistent with international LCA evidence and indicating that our data capture meaningful country-specific differences in production and logistics (*Al-Masbhi et al., 2026*).

2.4. Carbon Tax and Inequality

To examine the distributional consequences of food carbon taxation, we adapt a framework commonly used in the analysis of indirect taxation (*Kakwani, 1977; Pfähler, 1990*). This approach decomposes the redistributive effect of a tax into two key components: tax progressivity and tax intensity, allowing us to assess whether a food carbon tax disproportionately affects lower-income households.

The redistributive impact is measured using the Reynolds–Smolensky (RS) index, which captures the change in income inequality induced by the tax:

$$RS = G_{\text{pre}} - CI_{\text{post, ranked by pre-tax income}} \quad (6)$$

where G_{pre} is the Gini coefficient of pre-tax income, and $CI_{\text{post, ranked by pre-tax income}}$ is the concentration index of post-tax income, ranked by pre-tax income. A negative RS indicates that the tax increases inequality (regressive), while a positive value indicates a progressive effect.

To evaluate progressivity, we compute the Kakwani index (K):

$$K = C_T - G_{\text{pre}} \quad (7)$$

Where C_T is the concentration index of the carbon tax burden. A negative Kakwani index suggests the tax burden falls more heavily on lower-income groups. The RS index can then be expressed as a function of the average tax rate and tax progressivity:

$$RS = \frac{\bar{t}}{1 - \bar{t}} \times K \quad (8)$$

Where $\bar{t}$ is the average tax rate (total tax paid divided by total pre tax income).

We assess the equity impact of a food carbon tax using LCA-based emissions data and calibrate the tax separately for each country. For each country, we compute the weighted mean equivalised disposable income and set a reference burden equal to 0.3% of this value. We then select the per-kilogram CO₂e tax rate that would generate this burden for a household with average income and average food-related emissions. The 0.3% level is modest in fiscal terms but large enough to reveal distributional patterns, and it aligns with earlier work that uses income-based benchmarks to compare environmental tax incidence across countries (Büchs and Schnepf, 2013; Dorband et al., 2019; Sterner, 2012). This country-specific rate is applied uniformly to all food emissions, meaning households with more carbon-intensive diets pay higher absolute amounts, and the tax-to-income ratio varies across the distribution. Post-tax income is calculated by deducting household-level carbon payments, and distributional effects are summarised using the Reynolds–Smolensky and Kakwani indices, which capture both tax intensity and progressivity.

2.5. Household vs Individual Surveys and Eurostat vs National HBS

HBS and Individual Dietary Surveys (IDS) provide complementary but distinct perspectives on food consumption. HBS record household-level food purchases over short periods, which allows analysis of spending patterns and diet costs but does not capture actual intake, food waste or intra-household distribution (Trichopoulou et al., 2002; Zintzaras et al., 1997). IDS, in contrast, use methods such as 24-hour recalls and food diaries to collect individual-level nutrient intake data. While they provide richer information on dietary adequacy and health, IDS are less widely available and are often not harmonised across countries (de Oliveira et al., 2019; Serra-Majem et al., 2003). In this study, all empirical analysis is based solely on HBS microdata; IDS are discussed only to clarify the differences between possible data sources and are not used in the construction of the dataset.

Similar trade-offs arise when comparing Eurostat's harmonised HBS with national-level HBS. Eurostat applies standardised COICOP classifications and weighting methods to facilitate cross-country comparison (Eurostat, 2020), but its food categories, typically 60–80 groups, reduce item-level detail and can mask cultural and dietary heterogeneity. National HBS surveys often record 200–300 food items, which supports finer dietary analysis and closer alignment with emissions and nutrition databases, but they differ in classification schemes, recall periods and sampling designs (Sekula et al., 2005; Zezza et al., 2017).

2.6. Harmonization of Food Categories and Conversion of Expenditures to Volume

This study uses microdata from Eurostat's HBS and national HBS datasets for seven European countries and Türkiye. National HBS datasets were collected for Türkiye and UK. These surveys provide detailed household-level food expenditure records across 70–100 food categories, depending on the country. To align expenditure data across countries and years, we collected country-specific average retail prices in 2024 from national statistical agencies and major supermarket chains. These prices were adjusted to match each survey year using the Consumer Price Index (CPI) at the food-category level, following the formula:

$$Price_{(2015, i)} = Price_{(2024, i)} / (CPI_{(2024, i)} / CPI_{(2015, i)}) \quad (9)$$

where $Price_i$ is the current price of food item i and CPI aligns prices to the household expenditure year.

Converting household expenditures into physical quantities enables integration of nutritional and environmental assessments, a common practice when direct consumption data are unavailable (Thompson and Subar, 2017). Several studies have adopted this method: Ecker and Qaim (2011) used market prices to convert Ghanaian HBS expenditures into quantities for diet quality analysis; Darmon et al. (2003) and Giskes et al. (2006) applied food price indices to adjust for regional and temporal price variations; and Monsivais and Drewnowski (2007) demonstrated the approach in high-income contexts. Despite its limitations, such as not capturing food waste, stockpiling, or intra-household distribution (Gibson and Rozelle, 2003), this conversion remains a pragmatic solution for large-scale, multi-country studies. Estimated quantities were calculated as:

$$V_{ij} = \frac{E_{ij}}{P_j} \quad (10)$$

where E_{ij} is the household expenditure on item j , and P_j is the adjusted price per kilogram.

To harmonise food items across heterogeneous datasets (HBS with around 70 items and LCA with around 90 items), we developed a seven-group classification that is common to all countries and maps cleanly onto both sources. The groups are present in all diets and differ clearly in carbon intensity and dietary role:

1. Grains, bread, cereal products and processed foods
2. Ruminant meat
3. Non-ruminant meat
4. Fish and seafood
5. Dairy and eggs
6. Fruits and vegetables
7. Beverages.

This structure allows consistent integration of expenditure, quantity and emissions data across countries. Harmonisation was implemented using Python rule-matching routines, followed by manual checks to align HBS items with the corresponding LCA categories.

For environmental assessment, we applied LCA-derived carbon intensities (kg CO₂e/kg) from Poore and Nemecek (2018) which synthesize emissions data from 570 studies covering 38,000 farms and 40 food products globally, together with country-level emission factors from Big Climate Database.²

Household-level quantities were aggregated by food group g :

$$Expenditure_{ig} = \sum_{j \in C_g} E_{ij} \quad (11)$$

$$TotalVolume_i = \sum_{g=1}^7 Volume_{ig} \quad (12)$$

This structured approach allows us to estimate food expenditure shares, carbon footprints and nutrient availability in a consistent way across countries, forming a clear foundation for the cross-country distributional analysis that follows.

3. Results and Discussion

This section presents results from the microsimulation model for eight countries. Household food expenditures are converted into quantities and linked to LCA-based emissions and caloric data to estimate tax burdens, food budget shares and the carbon intensity of diets by income decile. We

2. The Big Climate Database (BCD) is a public, product-level environmental footprint database developed by the Danish Consumer Council (Forbrugerrådet Tænk) in collaboration with leading LCA experts. <https://denstoreklimadatabase.dk/en>

then assess distributional effects using standard inequality and progressivity measures, including pre- and post-tax Gini coefficients, the Kakwani index and the Reynolds–Smolensky index. Finally, a sensitivity analysis examines how alternative LCA emission factors affect the pattern of results across food groups and countries.

3.1. Food Price Patterns

Energy-adjusted food prices provide a useful indication of cross-country differences in affordability and the potential impact of a food carbon tax. **Figure 1** shows the average price per 1,000 kJ for each food group. Consistent with earlier work on diet costs (*Darmon et al., 2003; Monsivais and Drewnowski, 2007*), fish and seafood are generally the most expensive sources of dietary energy, followed by non-ruminant meat and beverages, while grains remain the lowest-cost energy source. Dairy, eggs and fruits and vegetables fall in the middle of the distribution, though their levels vary across countries. These differences reflect cross-country variation in retail price structures, food preferences, and the types of foods most purchased in each category.

Figure 2 scales these energy-adjusted prices by GDP per capita to indicate how the cost of dietary energy relates to national income levels, following previous work on food affordability (*FAO, 2025; Reisch et al., 2013*). Once income differences are accounted for, the relative position of countries changes substantially. High-income countries such as Denmark, Luxembourg and Ireland, where households have greater purchasing power, display lower values on this indicator despite having high nominal food prices. By contrast, Poland and Türkiye show higher values, reflecting tighter food budgets relative to income. This combined measure should be interpreted as indicative rather than definitive, but it helps distinguish between food groups that are costly yet more affordable in high-income settings and those that may pose genuine financial pressure for households in lower-income contexts.

The variation observed across **Figures 1 and 2** suggests that the same tax-induced price change could translate into different levels of financial pressure across countries, depending on local purchasing power and the role of specific foods in national diets. In Poland and Türkiye, most food

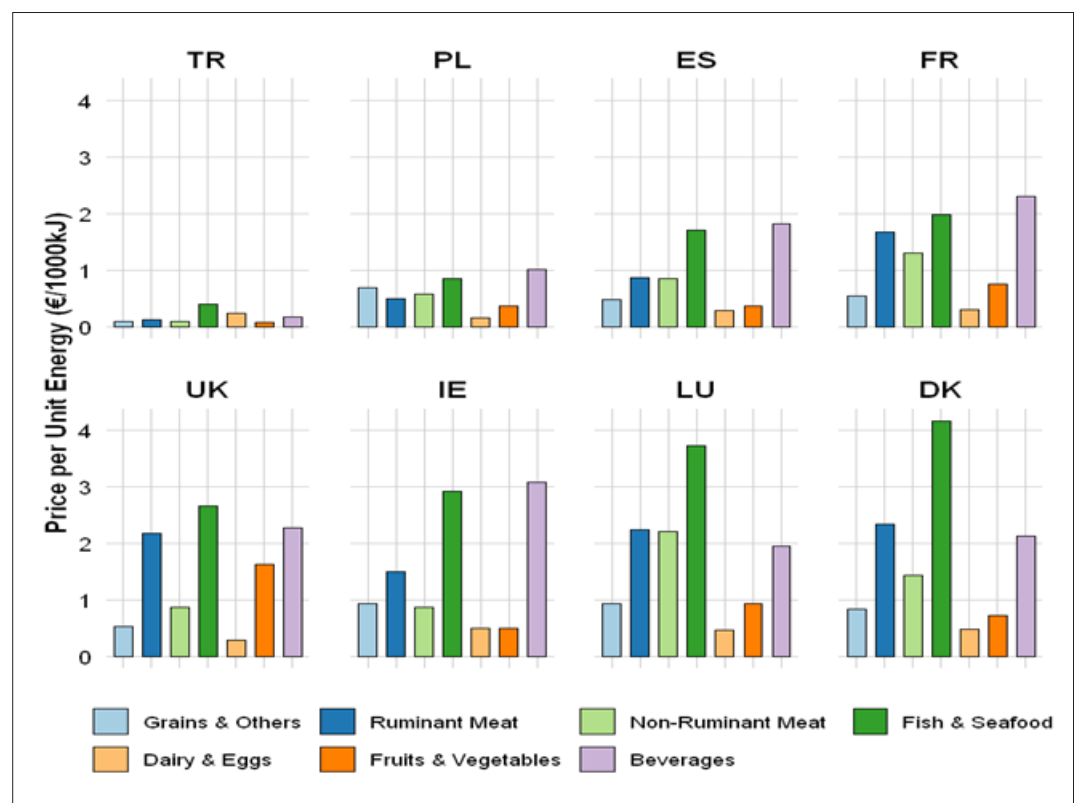


Figure 1. Price per 1000 kJ of Food Energy by Category and Country (€/1000kJ)

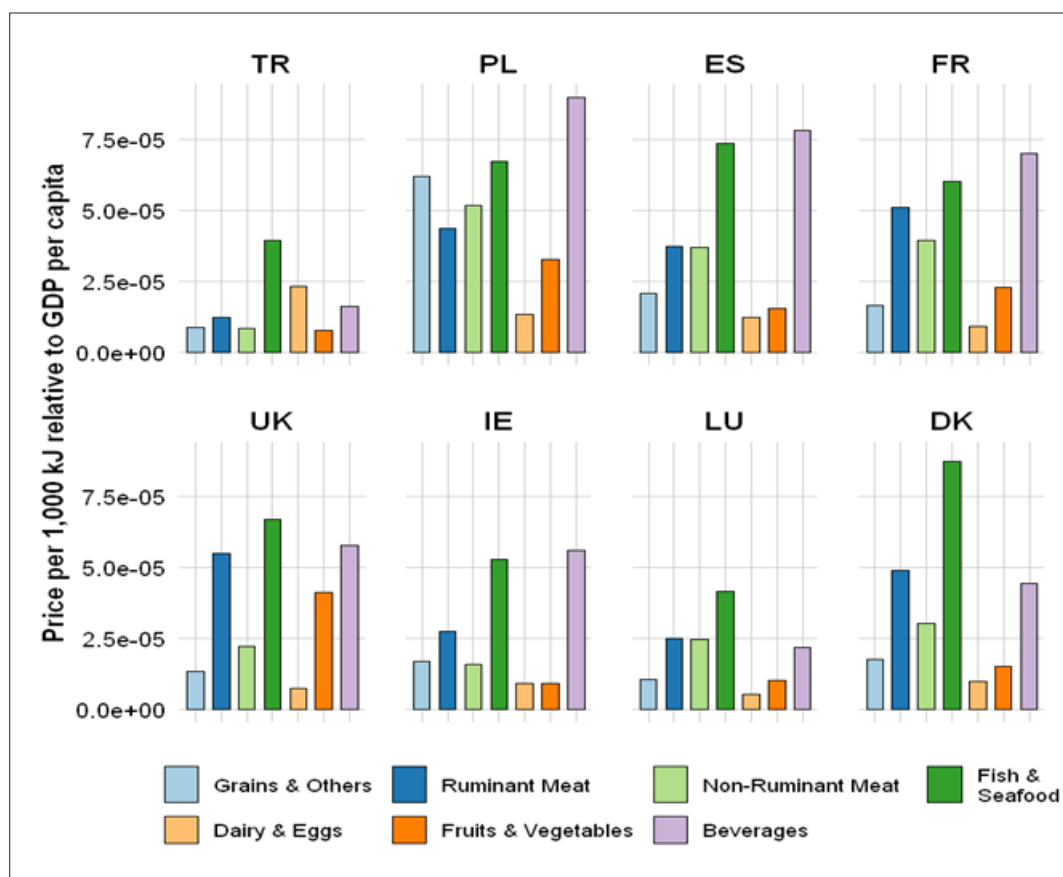


Figure 2. Price per unit of Energy by Food Group per Country (€/1000 kJ per GDP per Capita)

groups have higher GDP-adjusted prices than in Denmark or Luxembourg, indicating that even modest price increases may be more difficult to absorb for households. By contrast, in high-income countries, higher nominal prices coexist with greater purchasing power, which can soften the immediate budget impact of a food carbon tax. Grains remain the most affordable source of dietary energy in all countries, although their budget share varies across the income distribution.

3.2. Expenditure and Consumption Pattern

Understanding how households allocate their food budgets across income groups is important for evaluating the distributional effects of food carbon pricing. **Figure 3** shows the composition of household food expenditure, expressed as a share of total food spending, across income deciles in the eight countries. In most cases, lower-income households devote a larger share of their food budget to staples such as grains, bread and cereals, which is in line with Engel's Law and reflects the importance of low-cost, energy-dense foods for meeting basic needs (*Anker, 2011*). Higher-income households tend to spend relatively more on discretionary categories, including ruminant meat, beverages and fish, which are typically more expensive and more carbon intensive, and in several countries the share of fruits and vegetables also rises with income, suggesting a shift towards more varied and higher-quality diets.

There are, however, some clear exceptions to the standard Engel pattern. In Türkiye, the budget share for grains and related products increases with income. This may reflect a shift from inexpensive homemade staples, such as unpackaged bread and bulk flour, towards more costly packaged breads, breakfast cereals and processed grain products among higher-income households. In Luxembourg, staple foods are relatively inexpensive compared with income, so their budget share remains almost constant across deciles, while in Denmark and Ireland the gradient is weak because core staples such as rye bread and oats are widely consumed across the income distribution. These cases illustrate

how cultural food habits and relative price levels can modify the usual income gradient in staple consumption.

These cross-country trends carry important implications for modelling food-sector carbon taxation. The strong income gradients in discretionary and emissions-intensive categories, such as ruminant meat and beverages, indicate higher price elasticity among wealthier households, offering greater mitigation potential with minimal nutritional risk (Bouyssou et al., 2024). Conversely, staples like grains and non-ruminant meats exhibit relatively flat budget shares across deciles, suggesting limited substitution potential and raising equity concerns if carbon taxes disproportionately impact these categories (García-Muros et al., 2017). Incorporating this heterogeneity in food composition is vital for microsimulation frameworks aiming to assess the nutritional, environmental, and affordability trade-offs of carbon pricing. Without such disaggregation, models risk oversimplifying household responses and misrepresenting both the regressivity and effectiveness of fiscal interventions targeting food systems.

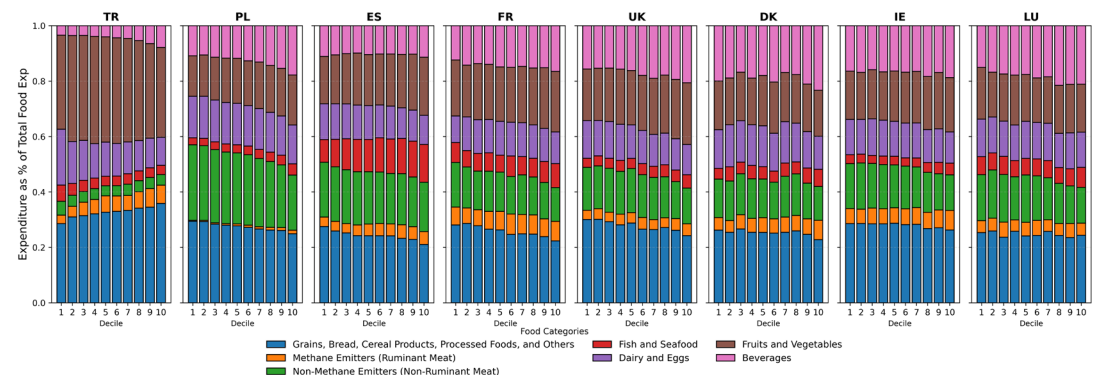


Figure 3. Composition of Food Expenditure by Category Across Income Deciles and Countries

3.3. Nutritional Implications: carbon intensity of dietary energy by income decile

As governments explore fiscal tools to address food system emissions, understanding who bears the burden of carbon pricing is essential for designing equitable policy. Figure 4 illustrates the energy-based carbon intensity of household food consumption, expressed relative to the national mean, across income deciles in the eight countries. This measure captures average greenhouse gas emissions per kilojoule (kJ) consumed, enabling cross-country comparison of dietary emission patterns by income level (Bonnet et al., 2018; Springmann et al., 2018).

A clear upward gradient is evident in France, Spain, Türkiye, Luxembourg, and the United Kingdom, where higher-income households consume diets with significantly higher emissions per unit of energy. These patterns are consistent with greater consumption of red meat, dairy, and processed foods among wealthier groups (Crippa et al., 2021; Poore and Nemecek, 2018). In contrast, lower-income households typically rely on more affordable, plant-based staples, which are both lower in emissions and cost. Ireland shows a non-linear pattern, with middle-income deciles exhibiting the highest carbon intensity. Denmark displays a much flatter profile, punctuated by a few noticeable spikes. These spikes appear to reflect data features rather than substantive dietary differences. The Danish HBS has a relatively small sample size compared with the other countries, so dividing the data into ten deciles amplifies the influence of a small number of households with unusually high purchases of ruminant meat or cheese (Deaton, 1997). This produces sharp movements between adjacent deciles that are unlikely to reflect systematic behavioural patterns. The overall picture remains one of relatively uniform diets across the income distribution, which is consistent with Denmark's comparatively compressed income structure (Causa et al., 2016; Heckman and Landersø, 2021). These results align with previous evidence on socio-nutritional gradients in Europe (Caillavet et al., 2019; Moberg et al., 2021) and illustrate how both dietary composition and underlying data structure can shape observed patterns.

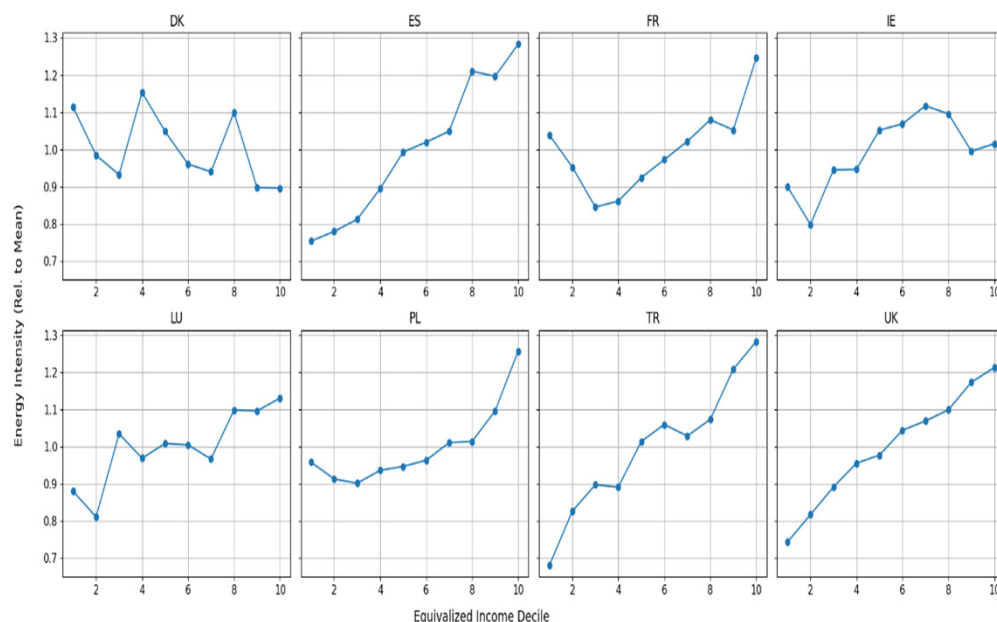


Figure 4. Energy-Based Carbon Intensity of Food Consumption Across Income Deciles in Eight European Countries (Relative to National Mean)

3.4. Carbon Tax Burden

Modelling the distributional effects of food-sector carbon taxation requires careful attention to both inter- and intra-income group disparities. **Figure 5** illustrates the carbon tax burden as a share of equivalized disposable income across deciles for eight countries, with boxplots highlighting within-decile variability. To facilitate cross-country comparison, the carbon tax is calibrated to a common reference level of 0.3% of average equivalized disposable income in each country. For every country, we first compute the weighted mean equivalized household income and take 0.3% of this value as the benchmark tax burden for the “average” household. Given baseline food-related emissions, we then choose a country-specific tax rate per kilogram of CO₂e that would generate this benchmark burden for a household with average income and average emissions. This rate is applied uniformly to all food emissions, so households with more carbon-intensive diets pay more in absolute terms, and the tax as a share of income varies across the distribution.

The analysis reveals a consistent regressive pattern across all countries: lower-income households bear a disproportionately higher tax burden relative to disposable income, while higher-income groups, despite contributing more in absolute terms, experience a lower percentage impact as shown in **Figure 5**. This finding aligns with prior studies on the regressivity of food and energy carbon taxes (*García-Muros et al., 2017; Revoredo-Giha et al., 2018*). The boxplots also reveal substantial variation within deciles, especially among lower-income households. This heterogeneity matters for distributional analysis: it shapes how households are affected by the tax and cautions against treating income groups as homogeneous. Ignoring these differences can lead to unrealistic assumptions about how households substitute between foods and may mask nutritional vulnerabilities within the same income bracket (*Garnett, 2016; Springmann et al., 2018*). These findings emphasize the importance of integrating granular household-level data into climate policy modelling frameworks to capture equity dimensions effectively.

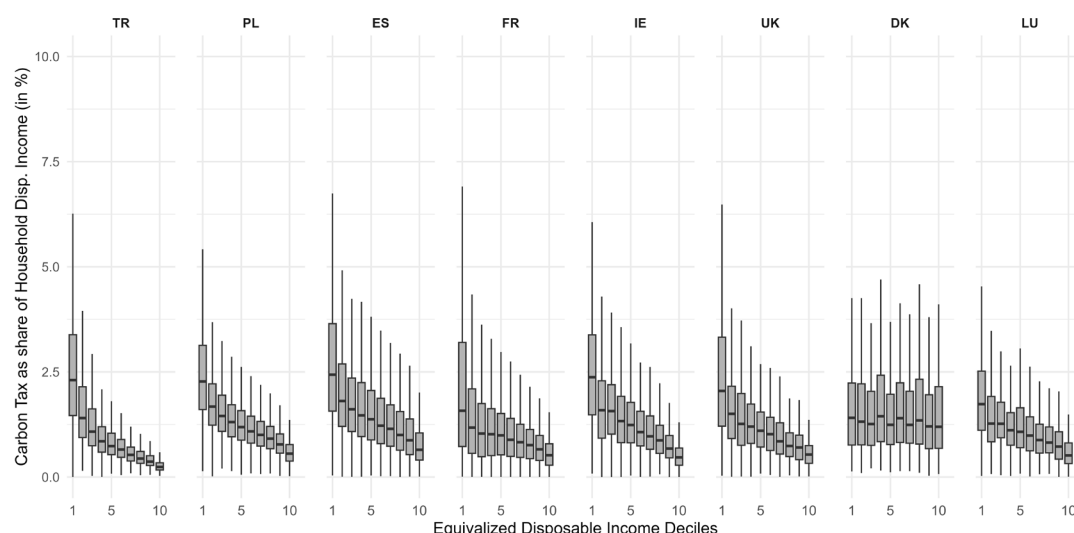


Figure 5. Carbon Tax Burden on Food as a Share of Income Across Income Deciles in Eight Countries

3.5. Distributional Impact of Carbon Tax

Assessing the distributional impact of a food carbon tax is especially important in a cross-country context because lower-income households tend to spend a larger share of their income on food and other carbon-intensive goods and are therefore more vulnerable to price increases (*García-Muros et al., 2017; Kehlbacher et al., 2016*). If these pressures are not offset by compensating measures, carbon pricing can reinforce existing income inequalities (*Piketty and Chancel, 2015*). To compare these effects across countries, we use standard inequality indicators, including the Gini coefficient, the Reynolds–Smolensky index and the Kakwani index, which summarise whether the tax burden is concentrated among poorer or richer households and whether the tax tends to reduce or increase overall inequality.

Table 1 shows that the food carbon tax has a modest but consistently regressive effect. In all countries, the post-tax Gini is slightly higher than the pre-tax value, and the Reynolds–Smolensky index is negative, indicating a small increase in income inequality. The Kakwani index is also negative everywhere, meaning that, relative to income, lower-income households bear a larger tax burden than richer households. The strength of this pattern differs across countries. Türkiye records the most negative Kakwani value, consistent with higher food budget shares and more limited scope to adjust spending. By contrast, Denmark has the least negative Kakwani index, reflecting the relatively flat pattern of food consumption across income deciles and higher baseline incomes. The results suggest that differences in food budget shares, dietary composition and the relative price of emission-intensive foods help explain why the same tax design produces different distributional outcomes across countries.

Table 1. Distributional Indicators of the Food Carbon Tax Across Countries

Country	Gini (Pre-Tax)	Gini (Post-Tax)	Reynolds–Smolensky Index	Kakwani Index	Average tax rate
IE	0.301	0.304	−0.0026	−0.2496	0.0103
FR	0.306	0.308	−0.0022	−0.1651	0.0122
DK	0.305	0.307	−0.0024	−0.1351	0.0163
UK	0.303	0.305	−0.0020	−0.2021	0.0098
TR	0.394	0.396	−0.0019	−0.3386	0.0055
LU	0.296	0.298	−0.0019	−0.1961	0.0096
PL	0.277	0.279	−0.0023	−0.2174	0.0105
ES	0.285	0.287	−0.0037	−0.1642	0.0129

These results indicate that even small food carbon taxes place a relatively larger burden on lower-income households, but that the sources of regressivity differ by country. Where regressivity is mainly driven by high food budget shares, income-based compensation may be most appropriate, while in countries where it reflects consumption of specific high-emission foods, targeted subsidies for low-carbon alternatives may be more effective. The cross-country variation suggests that a uniform European tax would create uneven impacts unless paired with country-specific support. The distributional indices therefore point to the need for future work that decomposes the tax burden into its underlying channels so that policy responses can be tailored to the particular drivers of regressivity in each setting.

3.6. Sensitivity Analysis

In the sensitivity analysis, we assess how dependent the results are on the choice of emission factors by comparing item specific LCA coefficients across Denmark, Spain, France, the United Kingdom, and a benchmark global average series. The confidence interval table indicates that for many staple food categories such as rice, flours and cereals, bread, pasta, fruit, vegetables, potatoes, oils, and most dairy products, the estimated means are accompanied by relatively narrow confidence intervals and low dispersion across countries. This suggests that within Europe the carbon intensity of these foods is fairly consistent and unlikely to drive major differences in downstream results. In contrast, several categories exhibit substantially wider confidence intervals and higher variability, most notably beef and veal, lamb and goat, some seafood categories, particularly frozen seafood, and a range of processed or composite foods such as crisps, dried fruit and nuts, confectionery products, and ready made meals. These patterns reflect heterogeneity in production systems, processing intensity, and methodological choices across LCA sources. Importantly, despite this variability, the overall ranking of food categories remains stable across countries and aligns with broader LCA evidence, with animal based products consistently appearing among the highest emitting foods and plant based staples remaining among the lowest. Overall, the results suggest that using harmonised EU or global average emission factors is unlikely to alter the qualitative conclusions regarding relative carbon intensities or the direction of distributional effects, although greater uncertainty should be acknowledged for high emission and highly processed food categories, as documented in **Supplementary file 1** and Appendix A.

4. Conclusion

This study develops a harmonized microsimulation framework to assess the distributional and nutritional effects of food carbon taxation across seven EU Member States and Türkiye. By linking LCA emissions intensities with nationally representative HBS, the framework provides a detailed cross-country analysis of how food-sector carbon pricing affects households along the income distribution. The eight countries were selected to reflect a wide range of dietary patterns, income levels and food system structures: northern and western European countries tend to consume more emission-intensive animal products, whereas Türkiye and several southern and eastern Member States rely more on plant-based staples and have higher food budget shares. This diversity enables an assessment of how a common carbon tax interacts with different national contexts.

A key contribution of the study is harmonizing both data structures and dietary profiles across countries. HBS surveys differ in recall periods, survey length and item aggregation, and countries vary in cultural food preferences and food basket composition (*Peltner and Thiele (2021)*). The harmonization procedure produces a consistent seven-group food classification and a unified link between expenditures, quantities, emissions and dietary energy. This ensures that observed cross-country differences reflect genuine variation in diets and costs rather than survey artefacts. The use of caloric carbon intensity, measured as CO₂e per kilojoule, further strengthens comparison by linking emissions directly to the energy content of diets.

The results show moderate but consistent regressivity: lower-income households bear a larger relative burden because they devote a higher share of income to food and have limited capacity to adjust diets, while higher-income households are affected less in proportional terms. Cross-country variation is pronounced. Türkiye displays the most regressive pattern, consistent with its higher food budget shares and the erosion of real incomes caused by sustained inflation in recent years, both of which weaken households' ability to absorb price increases (*OECD, 2023*). In contrast, Denmark has the least negative Kakwani index, reflecting flatter food expenditure gradients across income

groups and the cushioning effect of a comprehensive welfare state and higher baseline purchasing power (**Andersen, 2025**). These structural differences help explain why countries with more emission-intensive diets show higher mitigation potential, whereas those with more plant-based consumption patterns experience smaller relative changes. They also suggest that in a context such as Denmark, where distributional impacts are comparatively mild and social protection systems are strong, the political and social feasibility of introducing agricultural or food-sector carbon pricing may be greater, as reflected in Denmark's plans to implement a tax on agricultural greenhouse gas emissions (**State of Green, 2024**). These findings highlight a clear trade-off between environmental effectiveness and distributional fairness. Taxing more emission-intensive foods can generate larger emissions reductions, but it also increases food costs and can be regressive where low-income households spend a large share of income on food or have limited capacity to adjust consumption. Overall, the cross-country differences highlight how dietary structures, socio-economic conditions and institutional contexts shape the equity implications of food carbon taxation.

The political feasibility of food carbon taxation also depends on perceived fairness and the visibility of compensating measures. Evidence on carbon pricing shows that public support increases when low- and middle-income households are clearly protected and when revenue recycling is transparent and easy to understand (**Carattini et al., 2018**). In this context, options such as lump-sum carbon dividends, targeted transfers to low-income households or subsidies for healthier, low-carbon foods can all help address equity concerns and strengthen acceptability. Framing food carbon taxes in terms of co-benefits for health and climate resilience has also been shown to improve public support (**Maestre-Andrés et al., 2019**).

The microsimulation is static: quantities are held constant, behavioural adjustments are not modelled, and the tax is assumed to be fully passed through into consumer prices. Empirical studies show that pass-through can be incomplete or over-shifted depending on market structure and policy shocks (**Erutku, 2019; Vercammen, 2025; Weyl and Fabinger, 2013**). Within these assumptions, the framework captures environmental impacts through changes in diet-related emissions and nutritional impacts through changes in the cost and carbon intensity of dietary energy. Because food is a basic necessity, these distributional and nutritional outcomes are closely linked to public acceptability: perceived unfairness can trigger resistance to carbon pricing even when mitigation benefits are substantial (**OECD, 2022**). Future work will extend the model by incorporating behavioural elasticities, substitution patterns and empirically grounded pass-through parameters, and by simulating alternative designs such as revenue recycling and targeted food subsidies. It will also decompose the channels through which food carbon taxes affect different income groups, so as to assess how different policy packages can reduce regressivity, account for country-specific drivers and improve acceptability.

ORCID iDs

Ghassan Al-Masbhi  <https://orcid.org/0000-0001-8538-9710>
 Cathal O'Donoghue  <https://orcid.org/0000-0003-3713-5366>
 Zeynep Gizem Can  <https://orcid.org/0000-0002-2852-4995>
 Anna Diop-Christensen  <https://orcid.org/0000-0001-9043-5195>
 Ana Montes  <https://orcid.org/0009-0008-0900-8603>
 Emily Paradowska  <https://orcid.org/>
 Martina Pezer  <https://orcid.org/0000-0003-3291-220X>
 Denisa Sologon  <https://orcid.org/0000-0002-0309-5952>

Acknowledgements

The authors would like to thank the anonymous reviewers for their constructive comments and valuable suggestions, which greatly improved the quality of this paper. We are also grateful to Jules Linden for his valuable feedback and support throughout the development of this work.

Funding

This work was conducted as part of the Protein-I project, kindly supported by the Department of Agriculture, Food and the Marine (DAFM), grant number 2021R546. Additional support was provided by the Croatian Science Foundation (HRZZ) under project MOBODL-2023-12-7190, funded by the

European Union – NextGenerationEU. This research was also supported by the Erasmus+ Programme of the European Union (ecoMOD Project, project number: 2023-1-LI01-KA220-HED-000157594).

Conflict of Interest

No competing interests reported.

Additional files

Supplementary files

- Supplementary file 1.

Disclosure

Generative AI tools were used only to support language editing and improve clarity during the preparation of this manuscript.

Data and code availability

This study uses restricted household budget survey microdata from Eurostat's HBS, TurkStat's Household Budget Survey for Türkiye, and the UK Living Costs and Food Survey accessed through the UK Data Service. These data cannot be publicly shared by the authors. The analytical code is available from the corresponding author upon reasonable request, subject to data access restrictions.

References

- Al-Masbhi G**, O'Donoghue C, Sologon DM. 2026. Food carbon taxation and distributional equity: Comparing LCA and IO approaches. *Cleaner Food Systems* **3**: 100020. DOI: <https://doi.org/10.1016/j.clfs.2026.100020>
- Al-Masbhi G**, O'Donoghue C, Sologon DM. 2026. Bridging Affordability, Nutrition, and Sustainability: A Systematic Review of Microsimulation Approaches in Food Security Research. *International Journal of Microsimulation* **19**: 33–59. DOI: <https://doi.org/10.34196/ijm.00334>
- Alves R**, Perelman J, Chang K, Millett C. 2024. Environmental impact of dietary patterns in 10 European countries; a cross-sectional analysis of nationally representative dietary surveys. *European Journal of Public Health* **34**: 992–1000. DOI: <https://doi.org/10.1093/eurpub/ckae088>
- Andersen TM**. 2025. *Migration and the Nordic Welfare Model*. CESifo working papers.
- Anker R**. 2011. Engel's law around the world 150 years later. *Political Economy Research Institute Working Paper* **247**.
- Aysoy C**, Kirli DH, Tumen S. 2015. How does a shorter supply chain affect pricing of fresh food? Evidence from a natural experiment. *Food Policy* **57**: 104–113. DOI: <https://doi.org/10.1016/j.foodpol.2015.10.003>
- Blonk H**, Kool A, Luske B, Waart S, Blonk Milieuadvies G, Vegetariërsbond N. 2008. *Environmental Effects of Protein-Rich Food Products in the Netherlands: Consequences of Animal Protein Substitutes*. Gouda (NL): Blonk Consultants. p. 1–19.
- Böhringer C**, Rosendahl KE, Storrøsten HB. 2017. Robust policies to mitigate carbon leakage. *Journal of Public Economics* **149**: 35–46. DOI: <https://doi.org/10.1016/j.jpubeco.2017.03.006>
- Bonnet C**, Bouamra-Mechemache Z, Corre T. 2018. An environmental tax towards more sustainable food: empirical evidence of the consumption of animal products in France. *Ecological Economics* **147**: 48–61. DOI: <https://doi.org/10.1016/j.ecolecon.2017.12.032>
- Bouyssou CG**, Jensen JD, Yu W. 2024. Food for thought: A meta-analysis of animal food demand elasticities across world regions. *Food Policy* **122**: 102581. DOI: <https://doi.org/10.1016/j.foodpol.2023.102581>
- Browne S**, Mullen A, Mulholland B, Lo C, Ruttledge A. 2023. Nutrition quality and food and packaging waste associated with the school food system: a pilot, citizen science study in an Irish secondary school. *Journal of Human Nutrition and Dietetics* **36**: 2310–2322. DOI: <https://doi.org/10.1111/jhn.13220>
- Büchs M**, Schnepf SV. 2013. Who emits most? Associations between socio-economic factors and UK households' home energy, transport, indirect and total CO2 emissions. *Ecological Economics* **90**: 114–123. DOI: <https://doi.org/10.1016/j.ecolecon.2013.03.007>
- Caillavet F**, Fadhuile A, Nichèle V. 2016. Taxing animal-based foods for sustainability: environmental, nutritional and social perspectives in France. *European Review of Agricultural Economics* **43**: 537–560. DOI: <https://doi.org/10.1093/erae/jbv041>
- Caillavet F**, Fadhuile A, Nichèle V. 2019. Assessing the distributional effects of carbon taxes on food: Inequalities and nutritional insights in France. *Ecological Economics* **163**: 20–31. DOI: <https://doi.org/10.1016/j.ecolecon.2019.04.020>
- Callan T**, Lyons S, Scott S, Tol RSJ, Verde S. 2009. The distributional implications of a carbon tax in Ireland. *Energy Policy* **37**: 407–412. DOI: <https://doi.org/10.1016/j.enpol.2008.08.034>

- Carattini S**, Carvalho M, Fankhauser S. 2018. Overcoming public resistance to carbon taxes. *Wiley Interdisciplinary Reviews. Climate Change* **9**: e531. DOI: <https://doi.org/10.1002/wcc.531>
- Causa O**, Hermansen M, Ruiz N, Klein C, Smidova Z. 2016. *Inequality in Denmark through the Looking Glass*.
- Cerutti AK**, Beccaro GL, Bruun S, Bosco S, Donno D, Notarnicola B, Bounous G. 2014. Life cycle assessment application in the fruit sector: State of the art and recommendations for environmental declarations of fruit products. *Journal of Cleaner Production* **73**: 125–135. DOI: <https://doi.org/10.1016/j.jclepro.2013.09.017>
- Cervigni R**, Dvorak I, Rogers JA. 2013. *Assessing Low-Carbon Development in Nigeria: An Analysis of Four Sectors*. World Bank Publications.
- Chaudhary A**, Gustafson D, Mathys A. 2018. Multi-indicator sustainability assessment of global food systems. *Nature Communications* **9**: 848. DOI: <https://doi.org/10.1038/s41467-018-03308-7>
- Clune S**, Crossin E, Verghese K. 2017. Systematic review of greenhouse gas emissions for different fresh food categories. *Journal of Cleaner Production* **140**: 766–783. DOI: <https://doi.org/10.1016/j.jclepro.2016.04.082>
- Crippa M**, Solazzo E, Guizzardi D, Monforti-Ferrario F, Tubiello FN, Leip A. 2021. Food systems are responsible for a third of global anthropogenic GHG emissions. *Nature Food* **2**: 198–209. DOI: <https://doi.org/10.1038/s43016-021-00225-9>
- Darmon N**, Ferguson E, Briend A. 2003. Do economic constraints encourage the selection of energy dense diets? *Appetite* **41**: 315–322. DOI: [https://doi.org/10.1016/s0195-6663\(03\)00113-2](https://doi.org/10.1016/s0195-6663(03)00113-2)
- Deaton A**, Irish M. 1984. Statistical models for zero expenditures in household budgets. *Journal of Public Economics* **23**: 59–80. DOI: [https://doi.org/10.1016/0047-2727\(84\)90067-7](https://doi.org/10.1016/0047-2727(84)90067-7)
- Deaton A**. 1997. *The Analysis of Household Surveys: A Microeconomic Approach to Development Policy*. World Bank Publications.
- Decoster A**, Loughrey J, O'Donoghue C, Verwerf D. 2010. How regressive are indirect taxes? a microsimulation analysis for five European countries. *Journal of Policy Analysis and Management* **29**: 326–350. DOI: <https://doi.org/10.1002/pam.20494>
- de Oliveira D**, de Moura Souza A, Levy RB, Sichieri R, Verly-Jr E. 2019. Comparison between household food purchase and individual food consumption in Brazil. *Public Health Nutrition* **22**: 841–847. DOI: <https://doi.org/10.1017/S1368980018002987>
- Dokova KG**, Pancheva RZ, Usheva NV, Haralanova GA, Nikolova SP, Kostadinova TI, Egea Rodrigues C, Singh J, Illner A-K, Aleksandrova K. 2022. Nutrition transition in Europe: East-West dimensions in the last 30 years—a narrative review. *Frontiers in Nutrition* **9**: 919112. DOI: <https://doi.org/10.3389/fnut.2022.919112>
- Dorband II**, Jakob M, Kalkuhl M, Steckel JC. 2019. Poverty and distributional effects of carbon pricing in low- and middle-income countries – a global comparative analysis. *World Development* **115**: 246–257. DOI: <https://doi.org/10.1016/j.worlddev.2018.11.015>
- Drejerska N**, Sobczak-Malicka W. 2023. Nurturing sustainability and health: exploring the role of short supply chains in the evolution of food systems—the case of Poland. *Foods* **12**: 4171. DOI: <https://doi.org/10.3390/foods12224171>
- Ecker O**, Qaim M. 2011. Analyzing nutritional impacts of policies: An empirical study for malawi. *World Development* **39**: 412–428. DOI: <https://doi.org/10.1016/j.worlddev.2010.08.002>
- Edjabou LD**, Smed S. 2013. The effect of using consumption taxes on foods to promote climate friendly diets – the case of Denmark. *Food Policy* **39**: 84–96. DOI: <https://doi.org/10.1016/j.foodpol.2012.12.004>
- Erutku C**. 2019. Carbon pricing pass-through: evidence from Ontario and Quebec's wholesale gasoline markets. *Energy Policy* **132**: 106–112. DOI: <https://doi.org/10.1016/j.enpol.2019.05.026>
- Eurostat**. 2020. Household consumption by purpose – COICOP. https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Household_consumption_by_purpose
- FAO**. 2025. The State of Food Security and Nutrition in the World (SOFI). <https://openknowledge.fao.org/items/ea9cebff-306c-49b7-8865-2aef3bfd25e2>
- Figari F**, Paulus A, Sutherland H. 2015. Microsimulation and policy analysis. *Handbook of Income Distribution*. Elsevier. p. 2141–2221. DOI: <https://doi.org/10.1016/B978-0-444-59429-7.00025-X>
- Forero-Cantor G**, Ribal J, Sanjuán N. 2020. Levying carbon footprint taxes on animal-sourced foods: a case study in Spain. *Journal of Cleaner Production* **243**: 118668. DOI: <https://doi.org/10.1016/j.jclepro.2019.118668>
- García-Muros X**, Markandya A, Romero-Jordán D, González-Eguino M. 2017. The distributional effects of carbon-based food taxes. *Journal of Cleaner Production* **140**: 996–1006. DOI: <https://doi.org/10.1016/j.jclepro.2016.05.171>
- Garnett T**. 2011. Where are the best opportunities for reducing greenhouse gas emissions in the food system (including the food chain)? *Food Policy* **36**: S23–S32. DOI: <https://doi.org/10.1016/j.foodpol.2010.10.010>
- Garnett T**. 2016. Plating up solutions. *Science* **353**: 1202–1204. DOI: <https://doi.org/10.1126/science.aah4765>
- Gibson J**, Rozelle S. 2003. Poverty and access to roads in Papua New Guinea. *Economic Development and Cultural Change* **52**: 159–185. DOI: <https://doi.org/10.1086/380424>
- Giskes K**, Turrell G, van Lenthe FJ, Brug J, Mackenbach JP. 2006. A multilevel study of socio-economic inequalities in food choice behaviour and dietary intake among the Dutch population: the GLOBE study. *Public Health Nutrition* **9**: 75–83. DOI: <https://doi.org/10.1079/phn2005758>
- González AD**, Frostell B, Carlsson-Kanyama A. 2011. Protein efficiency per unit energy and per unit greenhouse gas emissions: potential contribution of diet choices to climate change mitigation. *Food Policy* **36**: 562–570. DOI: <https://doi.org/10.1016/j.foodpol.2011.07.003>
- Gumus SG**, Olgun FA, Adanacioglu H. 2010. Food consumption patterns in rural Turkey and poverty. *African Journal of Agricultural Research* **5**: 016–027.

- Heckman JJ, Landersø R.** 2021. *Lessons from Denmark about Inequality and Social Mobility*. National Bureau of Economic Research.
- Hyland JJ, Henschion M, McCarthy M, McCarthy SN.** 2017. The role of meat in strategies to achieve a sustainable diet lower in greenhouse gas emissions: a review. *Meat Science* **132**: 189–195. DOI: <https://doi.org/10.1016/j.meatsci.2017.04.014>
- Hynes S, Morrissey K, O'Donoghue C, Clarke G.** 2009. A spatial micro-simulation analysis of methane emissions from Irish agriculture. *Ecological Complexity* **6**: 135–146. DOI: <https://doi.org/10.1016/j.ecocom.2008.10.014>
- Ivanova D, Vita G, Steen-Olsen K, Stadler K, Melo PC, Wood R, Hertwich EG.** 2017. Mapping the carbon footprint of EU regions. *Environmental Research Letters* **12**: 054013. DOI: <https://doi.org/10.1088/1748-9326/aa6da9>
- Kakwani NC.** 1977. Measurement of tax progressivity: an international comparison. *The Economic Journal* **87**: 71–80. DOI: <https://doi.org/10.2307/2231833>
- Kebreab E, Mitloehner F, Sumner D.** 2022. Meeting the call: How California is pioneering a pathway to significant dairy sector methane reduction. UC Davis CLEAR Center <https://clear.ucdavis.edu/sites/g/files/Dgvnsk7876/files/inline-files/Meeting-the-Call-California-Pathway-to-Methane-Reduction.pdf> Archived at: <https://perma.cc/2AHE-SGAW>
- Kehlbacher A, Tiffin R, Briggs A, Berners-Lee M, Scarborough P.** 2016. The distributional and nutritional impacts and mitigation potential of emission-based food taxes in the UK. *Climatic Change* **137**: 121–141. DOI: <https://doi.org/10.1007/s10584-016-1673-6>
- Klenert D, Mattauch L, Combet E, Edenhofer O, Hepburn C, Rafaty R, Stern N.** 2018. Making carbon pricing work for citizens. *Nature Climate Change* **8**: 669–677. DOI: <https://doi.org/10.1038/s41558-018-0201-2>
- Maestre-Andrés S, Drews S, van den Bergh J.** 2019. Perceived fairness and public acceptability of carbon pricing: a review of the literature. *Climate Policy* **19**: 1186–1204. DOI: <https://doi.org/10.1080/14693062.2019.1639490>
- Moberg E, Säll S, Hansson P-A, Röö E.** 2021. Taxing food consumption to reduce environmental impacts – Identification of synergies and goal conflicts. *Food Policy* **101**: 102090. DOI: <https://doi.org/10.1016/j.foodpol.2021.102090>
- Monsivais P, Drewnowski A.** 2007. The rising cost of low-energy-density foods. *Journal of the American Dietetic Association* **107**: 2071–2076. DOI: <https://doi.org/10.1016/j.jada.2007.09.009>
- Murakami K, Livingstone MBE.** 2018. Greenhouse gas emissions of self-selected diets in the UK and their association with diet quality: is energy under-reporting a problem? *Nutrition Journal* **17**: 27. DOI: <https://doi.org/10.1186/s12937-018-0338-x>
- Notarnicola B, Tassielli G, Renzulli PA, Castellani V, Sala S.** 2017. Environmental impacts of food consumption in Europe. *Journal of Cleaner Production* **140**: 753–765. DOI: <https://doi.org/10.1016/j.jclepro.2016.06.080>
- Ntombela SM, Bohlmann HR, Kalaba MW.** 2019. Greening the South Africa's economy could benefit the food sector: evidence from a carbon tax policy assessment. *Environmental and Resource Economics* **74**: 891–910. DOI: <https://doi.org/10.1007/s10640-019-00352-9>
- O'Donoghue C.** 2014. *Handbook of Microsimulation Modelling*. Emerald Group Publishing.
- OECD.** 2022. Pricing Greenhouse Gas Emissions. https://www.oecd.org/en/publications/pricing-greenhouse-gas-emissions_e9778969-en.html
- OECD.** 2023. *OECD Economic Surveys: Türkiye 2023*. OECD Publishing, Paris. https://www.oecd.org/content/dam/oecd/en/publications/reports/2023/02/oecd-economic-surveys-turkiye-2023_816ce823/864ab2ba-en.pdf
- Peltner J, Thiele S.** 2021. Elasticities of Food Demand in Germany – A Demand System Analysis Using Disaggregated Household Scanner Data. *German Journal of Agricultural Economics* **70**: 49–62. DOI: <https://doi.org/10.30430/70.2021.1.49-62>
- Pfähler W.** 1990. Redistributive effect of income taxation: decomposing tax base and tax rates effects. *Bulletin of Economic Research* **42**: 121–129. DOI: <https://doi.org/10.1111/j.1467-8586.1990.tb00295.x>
- Piketty T, Chancel L.** 2015. *Carbon and Inequality: From Kyoto to Paris*. Trends in the Global Inequality of Carbon Emissions (1998-2013) and Prospects for An Equitable Adaptation Fund. Paris: Paris School of Economics.
- Poore J, Nemecek T.** 2018. Reducing food's environmental impacts through producers and consumers. *Science* **360**: 987–992. DOI: <https://doi.org/10.1126/science.aag0216>
- Ramstein C.** 2019. *State and Trends of Carbon Pricing 2019*.
- Reisch L, Eberle U, Lorek S.** 2013. Sustainable food consumption: an overview of contemporary issues and policies. *Sustainability: Science, Practice and Policy* **9**: 7–25.
- Rejman K, Halicka E, Kowrygo B.** 2014. Sustainable diet and changes in food consumption in chosen European Union Countries. *Zeszyty Naukowe SGGW w Warszawie - Problemy Rolnictwa Światowego* **14**: 132–140. DOI: <https://doi.org/10.22630/PRS.2014.14.4.78>
- Revoredo-Giha C, Chalmers N, Akaichi F.** 2018. *Measuring the Trade-off between Greenhouse Gas Emissions and Nutrition Due to Carbon Consumption Taxes in the UK*.
- Ricci M, Perez Dominguez I, Van Houtven S, Hristov J, Vandyck T.** 2024. *Pricing GHG Emissions in Agriculture: Accounting for Trade and Fairness for Effective Climate Policy*. JRC Working Papers on Taxation and Structural Reforms. DOI: <https://doi.org/10.2139/ssrn.5022621>
- Rippin HL, Cade JE, Berrang-Ford L, Benton TG, Hancock N, Greenwood DC.** 2021. Variations in greenhouse gas emissions of individual diets: associations between the greenhouse gas emissions and nutrient intake in the United Kingdom. *PloS One* **16**: e0259418. DOI: <https://doi.org/10.1371/journal.pone.0259418>

- Roxburgh N**, Burton RJF, Mittenzwei K, Polhill JG. 2025. Modelling the effect of a carbon tax on the development of a cultivated protein industry: a Norwegian case study. *Cleaner Engineering and Technology* **26**: 100979. DOI: <https://doi.org/10.1016/j.clet.2025.100979>
- Säll S**, Gren I-M. 2015. Effects of an environmental tax on meat and dairy consumption in Sweden. *Food Policy* **55**: 41–53. DOI: <https://doi.org/10.1016/j.foodpol.2015.05.008>
- Segal AB**, Olney J, Case KK, Sassi F. 2022. The benefits and challenges of taxing sugar in a small island state: an interrupted time series analysis. *The International Journal of Behavioral Nutrition and Physical Activity* **19**: 69. DOI: <https://doi.org/10.1186/s12966-022-01308-x>
- Sekula W**, Nelson M, Figurska K, Oltarzewski M, Weisell R, Szponar L. 2005. Comparison between household budget survey and 24-hour recall data in a nationally representative sample of Polish households. *Public Health Nutrition* **8**: 430–439. DOI: <https://doi.org/10.1079/phn2004695>
- Serra-Majem L**, MacLean D, Ribas L, Brulé D, Sekula W, Prattala R, Garcia-Closas R, Yngve A, Lalonde M, Petrasovits A. 2003. Comparative analysis of nutrition data from national, household, and individual levels: results from a WHO-CINDI collaborative project in Canada, Finland, Poland, and Spain. *Journal of Epidemiology and Community Health* **57**: 74–80. DOI: <https://doi.org/10.1136/jech.57.1.74>
- Shonkwiler JS**, Yen ST. 1999. Two-Step Estimation of a Censored System of Equations. *American Journal of Agricultural Economics* **81**: 972–982. DOI: <https://doi.org/10.2307/1244339>
- Smed S**, Scarborough P, Rayner M, Jensen JD. 2016. The effects of the Danish saturated fat tax on food and nutrient intake and modelled health outcomes: an econometric and comparative risk assessment evaluation. *European Journal of Clinical Nutrition* **70**: 681–686. DOI: <https://doi.org/10.1038/ejcn.2016.6>
- Smith P**, Gregory PJ. 2013. Climate change and sustainable food production. *The Proceedings of the Nutrition Society* **72**: 21–28. DOI: <https://doi.org/10.1017/S0029665112002832>
- Sørensen PB**, Beck UR, Berg AK, Christiansen S, Jørgensen CL, Kirk JS, Stewart LB, Stephensen PP. 2025. The effects of unilateral climate policy towards agriculture: a case study of Denmark. *Journal of Agricultural Economics* **76**: 3–23. DOI: <https://doi.org/10.1111/1477-9552.12624>
- Springmann M**, Sacks G, Ananthapavan J, Scarborough P. 2018. Carbon pricing of food in Australia: an analysis of the health, environmental and public finance impacts. *Australian and New Zealand Journal of Public Health* **42**: 523–529. DOI: <https://doi.org/10.1111/1753-6405.12830>
- State of Green**. 2024. Cutting agriculture emissions and restoring nature: Denmark's historic tripartite agreement. https://stateofgreen.com/en/news/cutting-agriculture-emissions-and-restoring-nature-discover-denmarks-historic-tripartite-agreement/?utm_source
- Sterner T**. 2012. Distributional effects of taxing transport fuel. *Energy Policy* **41**: 75–83. DOI: <https://doi.org/10.1016/j.enpol.2010.03.012>
- Tanton R**. 2014. A review of spatial microsimulation methods. *International Journal of Microsimulation* **7**: 4–25. DOI: <https://doi.org/10.34196/ijm.00092>
- Thompson FE**, Subar AF. 2017. Dietary assessment methodology. *Nutrition in the Prevention and Treatment of Disease* 5–48. DOI: <https://doi.org/10.1016/B978-0-12-802928-2.00001-1>
- Tiffin R**, Kehlbacher A, Salois M. 2015. The effects of a soft drink tax in the UK. *Health Economics* **24**: 583–600. DOI: <https://doi.org/10.1002/hec.3046>
- Trichopoulou A**, Naska A, Costacou T, Group DI. 2002. Disparities in food habits across Europe. *The Proceedings of the Nutrition Society* **61**: 553–558. DOI: <https://doi.org/10.1079/pns2002188>
- Trolle E**, Nordman M, Lassen AD, Colley TA, Mogensen L. 2022. Carbon footprint reduction by transitioning to a diet consistent with the Danish climate-friendly dietary guidelines: a comparison of different carbon footprint databases. *Foods* **11**: 1119. DOI: <https://doi.org/10.3390/foods11081119>
- Varacca A**, Castellari E, Moro D, Tiboldo G, Dogbe W, Gil JM, Revoredo-Giha C, Akaichi F, Irz X, Surry Y, Amuakwa-Mensah F, Sckokai P. 2024. Simulating the impact of a carbon tax on food in four European countries. *Q Open* **4**: qoae023. DOI: <https://doi.org/10.1093/qopen/qoae023>
- Vercammen J**. 2025. Carbon tax pass-through in western Canadian grain transportation. *Canadian Journal of Agricultural Economics/Revue Canadienne d'agroeconomie* **73**: 339–357. DOI: <https://doi.org/10.1111/cjag.70007>
- Vieux F**, Darmon N, Touazi D, Soler LG. 2012. Greenhouse gas emissions of self-selected individual diets in France: changing the diet structure or consuming less? *Ecological Economics* **75**: 91–101. DOI: <https://doi.org/10.1016/j.ecolecon.2012.01.003>
- Wadud Z**, Noland RB, Graham DJ. 2008. Equity analysis of personal tradable carbon permits for the road transport sector. *Environmental Science & Policy* **11**: 533–544. DOI: <https://doi.org/10.1016/j.envsci.2008.04.002>
- Weyl EG**, Fabinger M. 2013. Pass-through as an economic tool: principles of incidence under imperfect competition. *Journal of Political Economy* **121**: 528–583. DOI: <https://doi.org/10.1086/670401>
- Zeza A**, Carletto C, Fiedler JL, Gennari P, Jolliffe D. 2017. Food counts. Measuring food consumption and expenditures in household consumption and expenditure surveys (HCES). Introduction to the special issue. *Food Policy* **72**: 1–6. DOI: <https://doi.org/10.1016/j.foodpol.2017.08.007>
- Zhang Q**, Li B, Liu J-Y, Deng Y, Zhang R, Wu W, Geng Y. 2024. Assessing the distributional impacts of ambitious carbon pricing in China's agricultural sector. *Ecological Economics* **217**: 108082. DOI: <https://doi.org/10.1016/j.ecolecon.2023.108082>
- Zintzaras E**, Kanellou A, Trichopoulou A, Nelson M. 1997. The validity of household budget survey (HBS) data: estimation of individual food availability in an epidemiological context. *Journal of Human Nutrition and Dietetics* **10**: 53–62. DOI: <https://doi.org/10.1046/j.1365-277X.1997.00484.x>

Appendix A

This table summarises the sensitivity analysis of food specific emission factors using Denmark, Spain, France, and the United Kingdom, and benchmarks these values against the global average series used in the simulations. For each HBS food category, mean_other is the four-country European average emission coefficient, while IEEmissions reports the corresponding global average coefficient. The columns ci_low and ci_high report the 95 percent confidence interval around the mean using a normal approximation, and ci_low_t and ci_high_t report the corresponding interval using the t distribution to reflect the small number of country sources. The results show that many staple foods, including cereals, bread, pasta, fruit, vegetables, potatoes, oils, milk, yoghurt, and beverages, have relatively narrow confidence intervals, indicating broadly consistent emission factors across the four countries. In contrast, wider intervals are observed for high emission animal-based products such as beef and lamb, as well as for several processed or composite categories including crisps, dried fruit and nuts, confectionery products, and readymade meals, reflecting heterogeneity in production systems, processing intensity, and LCA assumptions. Despite this variation, the relative ordering of food categories by emissions remains stable, with animal-based products consistently exhibiting higher carbon intensities than plant-based staples, suggesting that the main qualitative conclusions are robust to plausible variation in the underlying emission coefficients. More details are found in the **Supplementary file 1**.

Table A1. Sensitivity analysis of food emission factors across EU countries

HBSVar	ci_low	ci_high	ci_low_t	ci_high_t	mean_other
Rice	3.585959	4.766041	3.340309	5.011692	4.27
Flours and other cereals	1.414113	1.857887	1.321735	1.950265	1.72
Bread	1.360303	1.639697	1.302143	1.697857	1.55
Other bakery products	1.766372	3.201628	1.467605	3.500395	2.78
Pizza and quiche	1.719267	2.932733	1.466668	3.185332	2.5825
Pasta products and couscous	1.770238	2.693762	1.577994	2.886006	2.465
Breakfast cereals	1.335546	2.396454	1.114703	2.617297	2.0075
Other cereal products	2.1078	8.556199	.7654805	9.89852	6.34
Beef and veal	50.19272	67.63927	46.56099	71.27101	62.945
Pork	4.759902	6.768098	4.341869	7.186131	5.38
Lamb and goat	27.81898	46.29702	23.97253	50.14347	40.2225
Poultry	1.821446	4.742554	1.213379	5.350621	2.5775
Other meats	3.205559	5.598442	2.707448	6.096552	3.8275
Edible offal	7.107489	20.13251	4.396157	22.84384	10.3
Dried, salted or smoked meat	5.832946	6.887054	5.613519	7.106481	6.275
Other meat preparations	1.9457	5.1023	1.288611	5.759388	2.73
Fresh or chilled fish	2.145666	3.406334	1.883241	3.668759	2.47
Frozen fish	2.226454	3.465546	1.96852	3.72348	2.5575
Fresh or chilled seafood	3.93944	4.47256	3.828464	4.583536	4.2575
Frozen seafood	4.910359	10.28564	3.791423	11.40458	8.4975

HBSVar	ci_low	ci_high	ci_low_t	ci_high_t	mean_other
Dried, smoked or salted fish and seafood	2.226108	5.133892	1.620814	5.739186	2.95
Other preserved or processed fish and seafood-based preparations	2.347288	6.180712	1.54931	6.97869	3.68
Milk, whole, fresh	-.0587203	2.35472	-.5611099	2.85711	.535
Milk, low fat, fresh	.1770255	2.490974	-.3046537	2.972654	.7675
Milk, preserved	.1464105	2.44159	-.3313615	2.919362	.7175
Yoghurt	.1569365	2.431064	-.3164532	2.904453	.7175
Cheese and curd	4.039816	6.724184	3.48103	7.28297	5.8275
Other milk products	.9050175	2.870983	.4957758	3.280224	1.46
Eggs	.4451219	3.118878	-.1114559	3.675456	1.1025
Butter	2.729667	3.402333	2.589643	3.542357	3.1325
Margarine and other vegetable fats	1.852835	4.647165	1.271158	5.228842	2.5375
Olive oil	5.791921	5.896079	5.77024	5.91776	5.83
Other edible oils	2.977505	4.842494	2.589283	5.230717	3.4375
Other edible animal fats	2.806462	5.373538	2.272091	5.907909	3.4375
Fresh or chilled fruit	.5109639	.7370361	.463904	.7840961	.605
Frozen fruit	.4104033	.6775967	.3547835	.7332165	.505
Dried fruit and nuts	4.800905	15.11909	2.653037	17.26696	12.275
Preserved fruit and fruit-based products	.8582021	1.185798	.7900087	1.253991	1.0775
Fresh or chilled vegetables other than potatoes and other tubers	.2552969	.4527031	.2142042	.4937958	.3175
Frozen vegetables other than potatoes and other tubers	.4497801	.6822199	.4013946	.7306053	.5825
Dried vegetables, other preserved or processed vegetables	1.347173	3.008827	1.001277	3.354723	2.5975
Potatoes	.4007628	.4632373	.3877579	.4762422	.44
Crisps	2.96352	5.68448	2.397116	6.250884	3.73
Other tubers and products of tuber vegetables	.4010697	.9629303	.284111	1.079889	.7525
Sugar	1.509345	2.726655	1.255946	2.980054	2.3975
Jams, marmalades and honey	1.304195	2.179805	1.121926	2.362074	1.9275
Chocolate	6.034944	14.36906	4.300088	16.10391	8.0775
Confectionery products	4.335207	22.39279	.5762805	26.15172	16.455

HBSVar	ci_low	ci_high	ci_low_t	ci_high_t	mean_other
Edible ices and ice cream	2.503622	3.308378	2.336101	3.475899	2.7325
Artificial sugar substitutes	1.109285	4.070715	.4928242	4.687176	2.9875
Sauces, condiments	1.918721	3.241278	1.643414	3.516586	2.7
Salt, spices and culinary herbs	.2951761	.6368238	.2240576	.7079424	.3825
Baby food	2.710623	2.801377	2.691731	2.820269	2.745
Ready-made meals	6.52434	30.62366	1.507746	35.64025	21.5175
Other food products n.e.c.	1.14756	5.11644	.3213852	5.942615	2.215
Coffee	16.47273	17.39527	16.28069	17.58731	17.0425
Tea	12.16886	25.57115	9.378991	28.36101	20.5625
Cocoa and powdered chocolate	16.64717	18.10083	16.34457	18.40343	17.0425
Mineral or spring waters	.109048	.198952	.0903333	.2176667	.1675
Soft drinks	.3924783	.9955217	.2669469	1.121053	.5425
Fruit and vegetable juices	1.391868	1.664132	1.335193	1.720807	1.585
Spirits and liqueurs	2.147991	3.800009	1.804101	4.143899	3.3925
Alcoholic soft drinks	.9878857	1.448114	.892083	1.543917	1.1975
Wine from grapes	1.531807	1.692193	1.498421	1.725579	1.615
Wine from other fruits	1.527807	1.688193	1.494421	1.721579	1.61
Fortified wines	1.449667	2.690333	1.191406	2.948594	2.1875
Wine-based drinks	1.722006	2.817994	1.493861	3.046139	2.4375
Lager beer	.5748086	.9371914	.4993738	1.012626	.67
Other alcoholic beer	.6128448	.9831551	.5357598	1.06024	.7225
Low and non-alcoholic beer	.6128448	.9831551	.5357598	1.06024	.7225
Beer-based drinks	.6128448	.9831551	.5357598	1.06024	.7225